

COMMENTS: EM-BASED LIKELIHOOD INFERENCE FOR SOME LIFETIME DISTRIBUTIONS BASED ON LEFT TRUNCATED AND RIGHT CENSORED DATA AND ASSOCIATED MODEL DISCRIMINATION

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This paper provides a comprehensive work on statistical estimation for lifetime data with left truncated and right censored observations. The authors derive several EM algorithm strategies (gradient EM, etc.) for the distributions commonly used in lifetime data analysis (from Weibull to generalized Gamma). All the algorithms are tested and experimented using simulation to prove their efficiency. The advantage of using a genuine EM algorithm as it is done by Balakrishnan and Mitra is that it has a provable ascent property for the observed log-likelihood, as any EM does. On the other hand, using an EM algorithm required the implementation of the M-step for the component parameters, which is specific to the parametric family involved, and may often be tedious, as illustrated by the calculations required in the present article.

To illustrate the alternative offered by Stochastic EM approaches, we focus here on the Weibull model alone. In the left truncated and right censoring setup the n lifetime data $\mathbf{t} = (t_1, \dots, t_n)$ iid $\sim f_\theta$ are associated with n truncation times $\boldsymbol{\tau} = (\tau_1, \dots, \tau_n)$ and n censoring times $\mathbf{c} = (c_1, \dots, c_n)$. The observed data is finally $(\tau, y, \delta) = ((\tau_1, y_1, \delta_1), \dots, (\tau_n, y_n, \delta_n))$, where $y_i = \min(t_i, c_i) \geq \tau_i$ and $\delta_i = \mathbb{I}_{\{t_i \leq c_i\}}$. The missing data are the censored observations $\{t_i, i \in \{1, \dots, n\} : d_i = 0\}$, and the complete data is $(\boldsymbol{\tau}, \mathbf{t})$ with complete log-likelihood function defined by

$$\ell^c(\theta | \boldsymbol{\tau}, \mathbf{t}) = \sum_{i=1}^n \{\log f_\theta(t_i) - \log(1 - F_\theta(\tau_i))\}, \quad (1)$$

where it is worth to note that $\tau_i = 0$ means that t_i is not left truncated.

When this is the case, Stochastic versions of EM like the one initially introduced by Celeux and Diebolt (1985) may overcome this difficulty at the expense of the loss of the ascent property, and more complicated convergence properties (general results from Nielsen (2000) give conditions for convergence of the sequence of estimates, which is generally a Markov Chain). The Stochastic EM (St-EM) principle consists of simulating the missing fraction of the data from its distribution

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conditional on the observed data, for the current value θ^k of the parameter. After this S-step, the estimation in the M-step gets back to standard MLE procedures applied to the complete data log-likelihood. In the present setup the conditional distribution of $(T|\tau, y, \delta)$ for the value θ of the parameter is the Dirac measure at $\{y\}$ when $\delta = 1$, and is when $\delta = 0$ the density of $(T|T > y, T > \tau)$ with $y > \tau$, that is

$$f_{\theta}(t|\tau, y, \delta = 0) = \frac{f_{\theta}(t)}{1 - F_{\theta}(y)} \mathbb{I}_{\{t \geq y\}}$$

where F_{θ} is the cdf associated with f_{θ} .

The iteration $\theta^k \rightarrow \theta^{k+1}$ of the St-EM algorithm is given by:

1. **E-step:** Calculate $f_{\theta^k}(t_i|\tau_i, y_i, \delta_i)$ for all $i = 1, \dots, n$;
2. **Stochastic step:** for $i : \delta_i = 0$, simulate $t_i^{k+1} \sim f_{\theta^k}(t|\tau_i, y_i, 0)$ to get completed data $\mathbf{t}^{k+1} = \{y_i : \delta_i = 1\} \cup \{t_i^{k+1} : \delta_i = 0\}$;
3. **M-step:** Calculate θ^{k+1} the MLE of θ based on $\ell^c(\theta|\tau, \mathbf{t}^{k+1})$ given by (1).

Several asymptotic results of ergodic averages from a St-EM algorithm are given by Nielsen (2000) under regularity assumptions. These results allow computation of asymptotic confidence intervals.

In the Weibull case $\theta = (\alpha, \eta)$ where α is the scale parameter and η is the shape parameter. The log-likelihood function is defined by

$$\ell(\theta|\tau, \mathbf{t}^{k+1}) = n \log \eta + (\eta - 1) \sum_{i=1}^n \log y_i^{k+1} - n \eta \log \alpha - \sum_{i=1}^n \frac{(y_i^{k+1})^{\eta} - \tau_i^{\eta}}{\alpha^{\eta}}.$$

For fixed η , the above function has an explicit maximizer with respect to α ; it is defined by

$$\alpha_n^{k+1}(\eta) = \left\{ \frac{1}{n} \sum_{i=1}^n \left[(y_i^{k+1})^{\eta} - \tau_i^{\eta} \right] \right\}^{1/\eta}.$$

Thus maximizing numerically the profiled log-likelihood defined by

$$\begin{aligned} & \eta \mapsto \ell^*(\eta|\tau, \mathbf{t}^{k+1}) \\ &= \ell((\alpha_n^{k+1}(\eta), \eta)|\tau, \mathbf{t}^{k+1}) \\ &= n(\log \eta - 1) + (\eta - 1) \sum_{i=1}^n \log y_i^{k+1} - n \log \left\{ \sum_{i=1}^n \left[(y_i^{k+1})^{\eta} - \tau_i^{\eta} \right] \right\} \end{aligned}$$

we obtain η_n^{k+1} and finally we set $\theta^{k+1} = (\alpha_n^{k+1}, \eta_n^{k+1})$ where $\alpha_n^{k+1} = \alpha_n^{k+1}(\eta_n^{k+1})$.

In our trials, at iteration $k + 1$ missing data are simulated by fixing θ to θ^k . Thus if $\delta_i = 0$ we have $t_i^{k+1} = F_{\theta^k}^{-1}(F_{\theta^k}(y_i) + u_i^{k+1}(1 - F_{\theta^k}(y_i)))$ where the u_i^{k+1} s are iid uniform on $(0, 1)$. The final estimate of θ is nothing but the ergodic mean $\bar{\theta}^K$ of the first K iterates where the variance part due to simulation is almost deleted. Then the asymptotic variance of the estimator can be derived following the method of Louis (1982) (see also formula (42) in Nielsen (2000)). As it is typically the case for EM algorithms, the choice of the starting parameter value $\theta^0 \in \Theta$ is important. On real data, a usual

Table 1: EM and St-EM estimates of the Weibull parameters for the left truncated and right censored data set provided by Balakrishnan and Mitra (2012). Confidence intervals computed from 1 000 non-parametric bootstrap replications are provided for the St-EM strategy.

Parameter	True	EM	St-EM	St-EM bootstrap 95% CI
α (scale)	34.813	34.364	34.313	(31.300, 37.608)
η (shape)	3.030	2.924	2.969	(2.412, 3.800)

practice consists in starting the algorithm from several values either taken from a grid, or randomly drawn from a uniform distribution on the parameter space, and retaining the EM estimate achieving the maximum of the observed likelihood among all the trials.

As an illustrative example we applied the St-EM method to the simulated data set provided in Balakrishnan and Mitra (2012). This data set of size 100 has 40 left truncated data and 50 right censored data. We choose the same initial value $\theta^0 = (28.2, 3.7)$ as in Balakrishnan and Mitra (2012). Contrary to Balakrishnan and Mitra we do not use the log-time transformation which leads to an extreme value distribution. The estimation results are given in Table 1 and show that St-EM estimates based on 10 000 iterations of the St-EM algorithm are very close to the results obtained by Balakrishnan and Mitra (2012). In addition, Table 1 shows 95% bootstrap percentiles obtained by running $B = 1000$ nonparametric bootstrap replications. For each bootstrap sample we computed initial values from moment estimates of the corresponding extreme value distribution, and ran 2 000 iterations of the St-EM algorithm.

One advantage of the Stochastic EM algorithm is that it can be extended easily to other models, such as parametric and even some semiparametric mixtures of lifetime distributions with censored data. See Bordes, Chauveau and Vandekerckhove (2007) and Bordes and Chauveau (2013). These algorithms are implemented in the mixtools package (Benaglia, Chauveau, Hunter and Young, 2009) for the R statistical software (R Development Core Team, 2010).

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